

Modeling Landscape-Level Spatial Variation in Sex Ratio Skew in the Mountain Pine Beetle (Coleoptera: Curculionidae)

Patrick M. A. James,^{1,2} Jasmine K. Janes,³ Amanda D. Roe,⁴ and Barry J. Cooke⁵

¹Département de Sciences Biologiques, Université de Montréal, Montréal, QC, Canada (patrick.ma.james@umontreal.ca),

²Corresponding author, e-mail: patrick.ma.james@umontreal.ca, ³Department of Biological Sciences, University of Alberta, Edmonton, AB, Canada (jasmine.janes@gmail.com), ⁴Canadian Forest Service, Great lakes Forestry Centre, Sault Ste. Marie, ON, Canada (amanda.roe@canada.ca), and ⁵Canadian Forest Service, Northern Forestry Centre, Edmonton, AB, Canada (barry.cooke@canada.ca)

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Abstract

Through their influence on effective population sizes, sex ratio skew affects population dynamics. We examined spatial variation in female-biased sex ratios in the mountain pine beetle (*Dendroctonus ponderosae* Hopkins) outbreak in western Canada to better understand how environmental context affects sex ratio skew. Our specific objectives were to: 1) characterize spatial variation in mountain pine beetle sex ratio; 2) test previously asserted hypotheses that beetle sex ratio varies with tree diameter and year in outbreak; and 3) develop predictive models of sex ratio skew for larval and adult populations. Using logistic regression, we modeled the probability that an individual beetle ($n = 2,369$) was female as a function of multiple environmental variables across 34 stands in British Columbia and Alberta, Canada. We identified a consistent female-biased sex ratio with significantly greater skew in adults (2:1, $n = 713$) than in larvae (1.76:1, $n = 1,643$). We found that the proportion of larval females increased with decreasing tree size and with outbreak age. However, adults did not respond to tree size and larvae did not respond to outbreak age. Predictive models differed between larvae and adults. All identified models perform well and included predictors related to weather, tree diameter, and year in outbreak. Female-biased sex ratios appear to originate from differential male mortality during development rather than from sex-biased oviposition, suggesting sex ratio skew is not the cause of outbreaks, but rather a consequence.

Key words: logistic regression, insect outbreak, spatial model, predictive modeling, forest ecology

Fisher predicted that there should be an equal investment in both sexes and that sex ratios within populations should be approximately even (Fisher 1930). If there is some benefit to being the rarer of the two sexes, then natural selection should result in a correction of such an imbalance. For example, if the reproductive cost of producing male offspring is less than that for females, and males are the rarer sex, then there is significant benefit in being and producing males. However, as the female:male ratio increases, there will be a selective shift toward the production of females, such that a 1:1 ratio is maintained. Hamilton (1967) expanded on these ideas and asserted that some variation could be expected as a result of variation in ploidy and sib-mating in wild populations. However, the general expectation over evolutionary time scales is that sex ratios should be ~1:1 in diploid systems.

Despite these theoretical predictions, many species exhibit strong variation from parity in sex ratio at shorter, ecological time scales (Zwölfer 1931, Utida 1947, Werren 1987, Craig et al. 1992, Mopper and Whitham 1992, Lachowsky and Reid 2014). This

phenomenon is particularly apparent in the spatio-temporal population dynamics of outbreaking insects and is important to consider due to its influence on effective population sizes and population growth rates.

Large variation in sex ratio has been shown to be associated with populations of outbreaking forest and agricultural insect pests. For example, the proportion of female European spruce beetle in Bavaria (*Ips typographus* L.) increased up to 72% during an outbreak and subsequently returned to a 1:1 ratio after the outbreak collapsed (Lobinger 1996). Similarly, in the spruce budworm (*Choristoneura fumiferana* Clem.), a population collapse was observed to be associated with a massive shift toward a male-biased sex ratio of 1:25 (Graham 1932). Scolytine beetles (Coleoptera: Curculionidae), such as the mountain pine beetle (*Dendroctonus ponderosae* Hopkins), generally exhibit female-biased sex ratios (Reid 1958, Cole et al. 1976, Amman and Cole 1983, Lachowsky and Reid 2014). The potential causes of such skew include differential mortality of males due to environmental stress (e.g., extreme

temperatures), demographic processes (e.g., maternal influences, thermal influence), or density-dependent male-killing endosymbiotic bacteria such as *Wolbachia* (Werren 1997). Increasing female bias is also thought to be associated with the evolutionary shift toward an increased parasitic lifestyle of bark beetles on their host trees (Lindgren and Raffa 2013). How these processes vary in space and time is uncertain, as is whether sex ratio skew is a cause or a consequence of insect population irruptions.

Mountain Pine Beetle

The mountain pine beetle is a native coleopteran pest of pine trees in western North America. Episodic outbreaks and their effects on forest structure and dynamics have significant broad-scale ecological and economic consequences (Safranyik et al. 2010). The mountain pine beetle is typically univoltine (Bentz et al. 2014), with pioneer females dispersing to new trees where they breed, construct a natal gallery, and oviposit if they are successful in overcoming the host tree's defences (Safranyik et al. 2010). During endemic periods, populations of mountain pine beetle are secondary tree colonizers (Costa 2006, Safranyik and Wilson 2006), only attacking weak or stressed trees. During epidemic periods, mountain pine beetle are considered primary colonizers and facultative predators (Lindgren and Raffa 2013) that mass attack and kill large mature trees through the combined effects of phloem girdling and beetle-vectored pathogenic ophiostomoid fungi (Lieutier et al. 2009, but see Six and Wingfield 2011). Further details on the transition from endemic to irruptive population dynamics can be found in Lindgren and Raffa (2013) and Raffa et al. (2015).

Recent changes in landscape-scale forest structure as a result of forest management and fire suppression, in combination with climatic warming, resulted in an unprecedented outbreak in Canada and the USA (Raffa et al. 2008, Bentz et al. 2010). Over 18×10^6 ha of forest have been affected in western Canada (Nealis and Cooke 2014), and over 8.9 million ha have been affected in the western United States (USDA Forest Service 2012). Continued northward expansion toward the Yukon territory, eastward expansion through Alberta toward Saskatchewan, and successful attack of jack pine (*Pinus banksiana*) in Alberta (Cullingham et al. 2011) pose a national-scale threat to the boreal forest ecosystem and further highlight the need for basic ecological understanding of this insect pest. In particular, understanding the drivers of spatial and temporal variation in sex ratio skew and the associated population dynamic consequences is needed to effectively monitor and manage populations (Robinson 1983).

Estimates of mountain pine beetle population growth rates depend on accurate estimates of effective population size (Bentz et al. 1993). An increase in the proportion of females increases effective population size and hence the rate of population growth. Currently, preventative approaches to forest management rely on estimates of a stand susceptibility index (SSI; Shore and Safranyik 1992, Björklund et al. 2008). An important component of this index is the r -value that estimates brood production per female (Bleiker et al. 2011). This value is calculated as the ratio of the combined number of live larvae, adults, and teneral adults to the number of entrance holes in a tree. In estimating brood production, the number of attacks per m^2 is uniformly multiplied by 1.6 to compensate for the assumed 2:1 female biased sex ratio (Reid 1958, Björklund et al. 2008). This approach may over- or underestimate population growth rates at particular locations if sex ratio is spatially and temporally variable. Understanding such variability in mountain pine beetle sex ratios,

and at what scales, over the course of outbreak development could provide important information for estimating stand susceptibility and managing beetle infestations.

To better understand sex ratio skew in mountain pine beetle populations, we examined at what life stage sex ratio skew arises (temporal dynamics) and which environmental factors predict the strength of this skew (spatial dynamics). Recently, Lachowsky and Reid (2014) showed that sex ratio skew in mountain pine beetle arises due to increased male mortality over the course of larval development and was not exclusively due to smaller male body-size. Lachowsky and Reid (2015) then showed that the skew toward female bias that arises in the larval stage is not attributable to male-killing endosymbiotic bacteria such as *Wolbachia*, suggesting that some other mechanism may be involved, such as the influence of not yet identified external environmental factors. It has previously been hypothesized that mountain pine beetle sex ratios vary in response to an interaction between tree size and year in outbreak (Fig. 1; Amman and Cole 1983). However, this hypothesis has not been formally tested and the actual role of these factors, the spatial variation in these effects, and the role of other spatial environmental factors on sex ratio variation remain uncertain.

In this paper, we explicitly test these previously described hypotheses across a broad geographic extent at the leading edge of an outbreak mountain pine beetle population in western Canada. Using logistic regression and an empirically validated sex-identifying microsatellite marker (Davis et al. 2009), we address three specific questions: 1) Can we find support for the generally accepted, but seldom tested, roles of tree size, climate, and stage of outbreak progression on mountain pine beetle sex ratio over a large geographic area? 2) Which other environmental factors can be used to predict sex ratio skew and improve population monitoring? and 3) How do the environmental determinants of sex ratio vary between larval and adult beetles?

Materials and Methods

Study Area and Data Collection

Larval and adult mountain pine beetle were collected continuously between September 2007 (postflight) and April 2008 (pre-emergence)

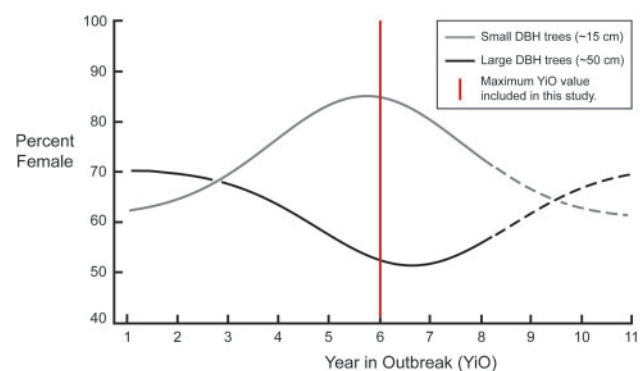


Fig. 1. Hypothesized effects of year in outbreak (YiO) and two extremes in tree diameter (DBH) on the proportion of female beetles (Fig. 17 from Amman and Cole 1983). Solid lines represent data taken from Wasatch-Cache National Forest, Utah, USA. Dashed lines represent trajectories postulated by the original authors. Our goal was to test the robustness of this hypotheses describing sex ratio variation using data from multiple sites across the leading edge of an mountain pine beetle outbreak for the portion of the curves to the left of the vertical red line (i.e., Year in Outbreak (YiO) = 0–6). Reprinted with permission from the USDA Forest Service.

(Fig. 2) from multiple attacked lodgepole pine trees (Cullingham et al. 2011) in multiple stands in Alberta and British Columbia, Canada, at what was the leading edge of the outbreak at that time (Fig. 3; Table 1). Sites were selected to represent a range of ecoregions and outbreak conditions across a portion of the range. A 10-cm-diameter hole-saw and chisel were used to extract 3-cm-deep pine discs composed of bark, xylem, and beetle-infested phloem. Collected adult beetles were placed in a Petri dish of agar for 24 h prior to processing to verify that they were alive. Tree diameter (DBH) was measured at a height of 1.3 m. The sex of each beetle was identified using a sex-identifying microsatellite marker (Davis et al. 2009). Further details on field collection and microsatellite genotyping are described in detail in Roe et al. (2010). A total of 2,369 beetles (842 male, 1,527 female / 1,644 larvae, 725 adults) were collected from 34 stands that included a total of 322 individual trees.

Statistical Analysis

To determine whether sex ratio differs between larvae and adult populations, we first compared the proportion of females in each life-stage across the study area using Fisher's exact test (Fisher 1935). We then tested whether sex ratio varied among sites (expressed as a categorical factor) using logistic regression model in which the sex of individual beetles (0 = male; 1 = female) was the response variable. Data describing all beetles (larvae + adults), only larvae, and only adults were examined. We next examined if geographic location (i.e., latitude and longitude) affected sex ratio variation. Beetles are sensitive to temperature and male beetles in particular are thought to be more susceptible to mortality due to desiccation and freezing (Amman and Cole 1983, Lachowsky and Reid 2014). One expects the relative number of females to increase over the progression of a season (i.e., fall → winter → spring) due to differential mortality. To examine the role of seasonal development on differential mortality, we also modelled beetle sex as a function

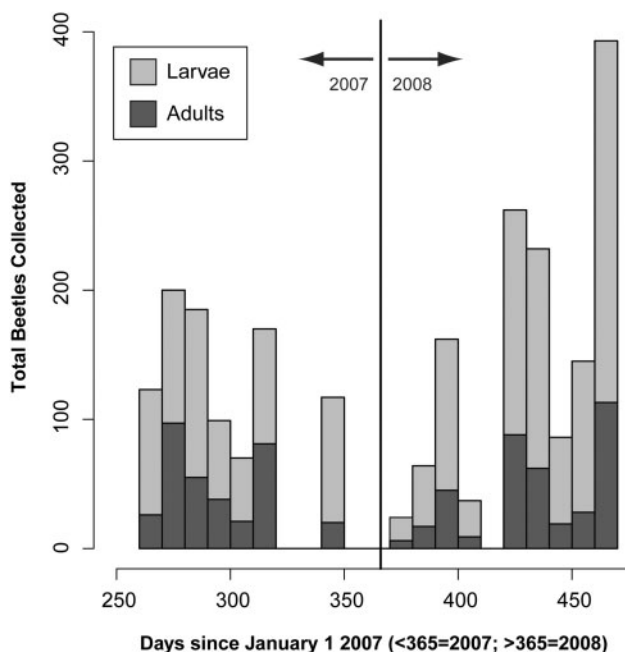


Fig. 2. Frequency histogram of the total number of beetles as well as the proportion of adult and larval females collected between 25 September 2007 and 12 April 2008. Live adult beetles were present throughout the sampling season.

of a continuous variable representing date of collection (days since January 1st, 2007; Fig. 2), once again using logistic regression.

Year in Outbreak

Year in outbreak (YiO) represents the number of years a site had been classified as attacked based on remotely sensed data on mountain pine beetle damage and attack in Alberta and British Columbia (Table 2). Data describing the extent and severity of beetle-caused forest mortality differ between Alberta and British Columbia. Mountain pine beetle damage in British Columbia is represented as polygons that are the product of fixed-wing aerial surveys of outbreak damage based on patches of crown discoloration. These data were converted to annual raster maps at a resolution of 2 km for analysis. In contrast, mountain pine beetle damage in Alberta is represented as points that are the product of helicopter surveys. To make the Alberta data compatible with the converted BC raster data, annual point layers were converted to raster using the Kernel Density function in ArcMap 10.1 (ESRI 2014). The number of trees affected at point locations in a given year served as the input data for the kernel density function. Rasters were created using a 2-km resolution and a 10-km kernel radius. This approach assumes that if beetles are found at a point, it is probable that beetles are also present in adjacent cells with exponentially decreasing probability with increasing distance, to a maximum distance of 10 km. Only pixels with a modelled probability of beetle occurrence of greater than 10% were considered "attacked." Raster layers were created for both jurisdictions for the years 1998–2008, inclusive. YiO was calculated as the number of times between 1998 and 2007/2008 (depending on collection date; Table 1) that a sample point fell in an attacked pixel.

Hypothesis Testing

Logistic regression was used to test the specific hypothesis that sex ratio varies in response to an interaction between tree size (DBH) and outbreak duration (YiO; Amman and Cole 1983; Fig. 1). As above, for all regression models, the sex of individual beetles (male / female) was used as the response variable. We first verified that the effects of our hypothesized covariates differed between beetle stage by testing for a significant interaction between beetle life stage (larva vs. adult) and our two principal covariates (DBH and YiO) in a model considering all beetles (larvae + adults). We then compared this model to others created for larvae and adults separately. All models, including those described below, were built using the *glm* function in R using the logit link-function.

Predictive Models

Statistical models were developed to predict beetle sex ratio skew using multiple environmental variables (Table 2) including tree diameter (DBH), year in outbreak (YiO), climate, weather, and geographic location. Our goal was to identify models that could be applied in new areas to predict sex ratio skew. Prior to analysis, all predictors were standardized to zero mean and unit variance. In our logistic regression approach, we assume that the probability of finding a female beetle within a tree serves as a proxy for overall tree-level sex ratio skew. Tree-level information (i.e., DBH) is the finest scale of data used in these models.

Predictive models for all beetles (larvae + adults), larvae, and adults were selected from sets of plausible candidate models (Table 3) using different combinations of the variables described above. This set of plausible models included a full model (all variables included) and an intercept-only model. Except for the

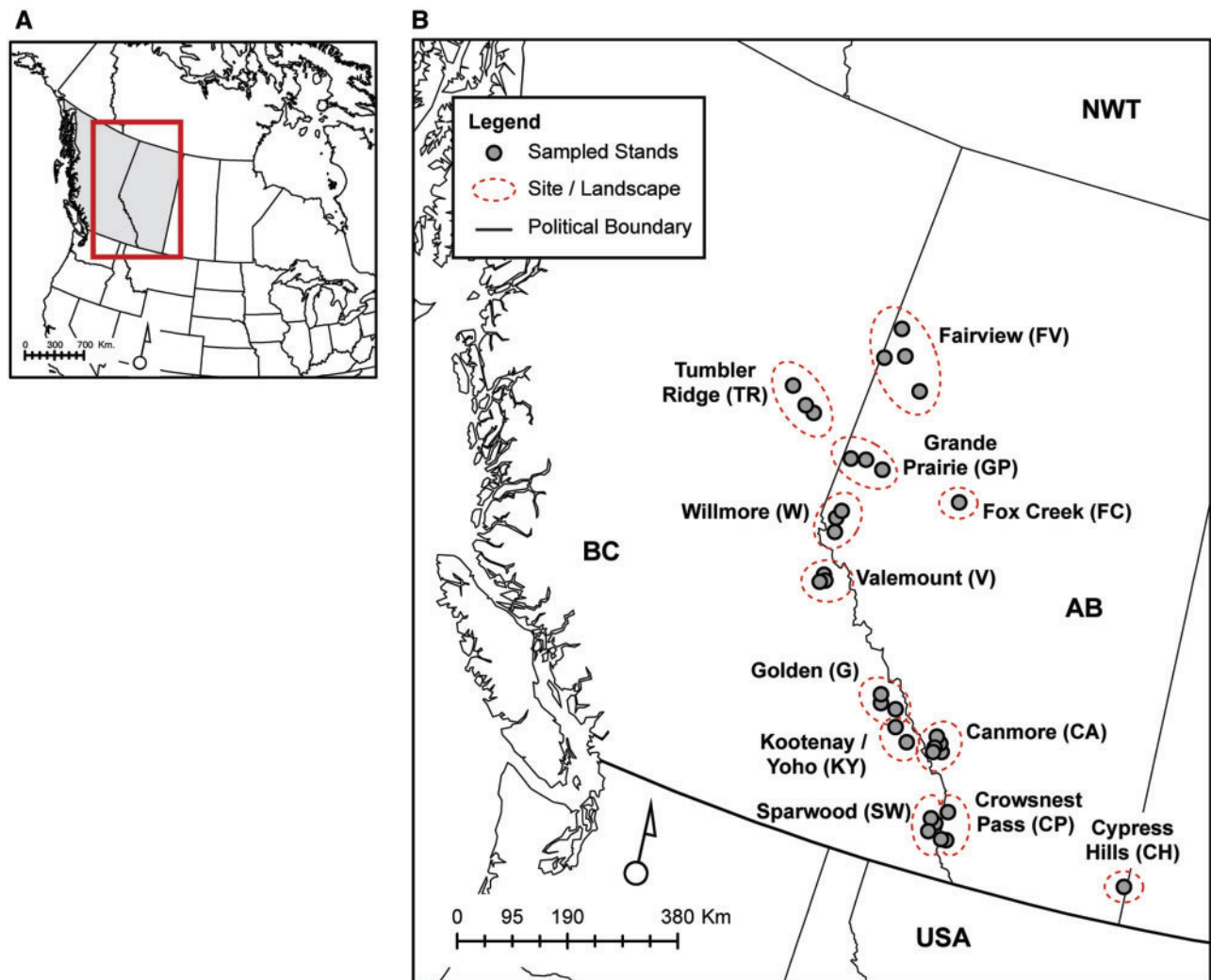


Fig. 3. (A) Continental context of study area, and (B) sampling locations (stands) in British Columbia and Alberta, Canada. Points represent the centroid of the minimum convex polygon for all of the trees used for a single stand. Site labels correspond with those indicated in Table 1.

intercept-only model, all models included DBH and YiO. We endeavoured to minimize correlations among variables. Weather variables related to seasonal temperature (Table 2) that exhibited a correlation > 0.7 were removed from the set of potential predictors. Our reduced set of temperature variables included: Maximum Summer, Maximum Fall, Minimum Winter, and Mean Spring. We consider this set of temperature variables to represent a parsimonious subset to serve as a proxy for the overall thermal regime at each site. Latitude was also correlated with Longitude ($r=0.88$) and Minimum Winter temperature ($r=0.82$); thus, when constructing our plausible models, Minimum Winter temperature and Latitude were not included together. When considering location alone, we chose to examine Latitude, as it represents the dominant axis of recent mountain pine beetle outbreak expansion (James et al. 2011, Samarasekera et al. 2012). We also included a model selected using a stepwise procedure based on Akaike's Information Criterion (AIC), without regard for correlation among predictors, for comparison. Stepwise model selection used the *stepAIC* function in the MASS package in R (Venables and Ripley 2013).

AIC corrected for small sample sizes (AIC_c) and Akaike weights were used to compare the 10 candidate models (Burnham and Anderson 2002). This approach seeks to identify the simplest and

most parsimonious model by maximizing the likelihood of the model given the data while minimizing the number of estimated parameters. The models with the greatest support given the data are those that minimize the AIC_c criterion. AIC_c values, log-likelihoods, and model weights were calculated using the *AICcmodavg* package in R (Mazerolle 2015).

Model Performance

Model performance was evaluated by comparing predicted and observed sex ratio (expressed as the percentage of female beetles) at the level of trees classified into six size-categories within sample sites. Although the predictive models were built using individuals, we could not assess their performance at this level as trees (the finest resolution in our data) contain both presences (females) and absences (males). Presences and absences in the same unit of analysis confound traditional methods for evaluating logistic regression model performance (e.g., pseudo- R^2 , ROC). Although the models cannot distinguish presences from absences, they can be used to predict expected tree-level sex ratio. Because individual trees often did not have enough surveyed individuals to provide a good estimate of the tree level sex ratio, trees within stands were grouped into six

Table 1. Details for each collection site including location and timing of collection; sites are plotted in Fig. 3

Site	Stand	Lat.	Long.	Date	Total beetles	Total $p(\varphi)$	Total larvae	Larval $p(\varphi)$	Total adults	Adult $p(\varphi)$
Canmore	CA.1	50.95	-115.34	26 Sept. 2007	66	0.61	50	0.58	16	0.69
	CA.2	50.92	-115.15	27 Sept. 2007	57	0.68	47	0.68	10	0.70
	CA.3	51.03	-115.24	11 Dec. 2007	68	0.75	48	0.73	20	0.80
	CA.4	50.88	-115.35	13 Dec. 2007	49	0.61	49	0.61	0	NA
	CA.5	51.12	-115.37	31 Jan. 2008	135	0.68	94	0.65	41	0.76
Cypress Hills	CH.1	49.60	-110.08	12 Oct. 2007	13	0.85	1	1.00	12	0.83
Crowsnest Pass	CN.1	49.66	-114.57	05 Feb. 2008	27	0.44	23	0.43	4	0.50
	CN.2	50.06	-114.58	07 Feb. 2008	22	0.59	16	0.63	6	0.50
	CN.3	49.63	-114.44	11 April 2008	23	0.52	19	0.47	4	0.75
Fox Creek	FC.1	54.71	-116.63	24 Jan. 2008	88	0.75	65	0.68	23	0.96
Fairview	FV.1	56.59	-119.37	14 Mar. 2008	69	0.55	60	0.57	9	0.44
	FV.2	56.16	-118.65	12 Mar. 2008	113	0.65	63	0.54	50	0.78
	FV.3	56.99	-119.74	13 Mar. 2008	50	0.70	47	0.70	3	0.67
	FV.4	56.46	-119.91	02 April 2008	57	0.56	48	0.56	9	0.56
Golden	G.1	51.08	-116.42	03 Oct. 2007	93	0.57	51	0.55	42	0.60
	G.2	51.36	-116.94	04 Oct. 2007	107	0.61	52	0.58	55	0.64
	G.3	51.50	-117.00	25 Oct. 2007	74	0.61	48	0.63	26	0.58
Grande Prairie	GP.1	54.81	-118.88	13 Feb. 2008	45	0.60	31	0.55	14	0.71
	GP.2	54.88	-119.40	28 Feb. 2008	125	0.66	86	0.65	39	0.67
	GP.3	54.80	-119.79	05 Mar. 2008	107	0.63	69	0.61	38	0.66
Kootenay / Yoho	KY.1	50.91	-116.04	08 April 2008	68	0.69	48	0.67	20	0.75
	KY.2	51.33	-116.54	08 April 2008	85	0.60	61	0.61	24	0.58
Sparwood	SW.1	49.68	-114.91	12 April 2008	71	0.70	48	0.69	23	0.74
	SW.2	49.84	-114.80	12 April 2008	81	0.64	48	0.60	33	0.70
	SW.3	49.89	-114.92	12 April 2008	65	0.62	56	0.63	9	0.56
Tumbler Ridge	TR.1	55.54	-121.98	06 Nov. 2007	70	0.61	49	0.67	21	0.48
	TR.3	55.32	-121.45	08 Nov. 2007	85	0.61	46	0.65	39	0.56
	TR.4	55.27	-121.18	09 Nov. 2007	85	0.72	43	0.77	42	0.67
Valemount	V.1	52.88	-119.30	16 Oct. 2007	49	0.63	49	0.63	0	NA
	V.2	52.85	-119.38	18 Oct. 2007	75	0.64	43	0.72	32	0.53
	V.3	52.96	-119.38	19 Oct. 2007	73	0.68	50	0.68	23	0.70
Willmore -Kakwa	WK.1	54.00	-119.52	28 Mar. 2008	59	0.66	48	0.63	11	0.82
	WK.2	53.65	-119.50	28 Mar. 2008	63	0.71	46	0.70	17	0.76
	WK.3	53.85	-119.59	27 Mar. 2008	52	0.67	42	0.67	10	0.70

Table 2. Summary of spatial variables tested for their ability to predict beetle sex ratio

Variable	Source	Format	Details	Mean (SD)
Geographic location	Field Data	Point	Latitude	/ 53.154 (2.223)
			Longitude	/ -117.868 (2.239)
Weather	BioSim (Régnière and St-Amant 2008)	Point data	Minimum	Spring -1.911 (1.870)
			Mean	Spring 4.230 (1.775)
			Maximum	Spring 10.370 (1.759)
			Minimum	Summer 6.310 (1.111)
			Mean	Summer 13.220 (1.077)
			Maximum	Summer 20.131 (1.708)
			Minimum	Fall -6.885 (1.382)
			Mean	Fall -2.091 (1.133)
			Maximum	Fall 2.702 (1.198)
			Minimum	Winter -13.733 (2.712)
			Mean	Winter -7.946 (2.191)
			Maximum	Winter -2.158 (1.929)
Climate	Climate WNA Tool. (Wang et al. 2012)	Point data	Mean Annual Temperature (MAT)	1.261 (1.172)
			Climate Moisture Deficit (CMD; a drought index).	117.1 (71.921)
			See text for details on calculation.	2.009 (1.797)
Years in Outbreak (YiO)	British Columbia Ministry of Forests & Alberta Agriculture and Forestry (formerly AERSD)	Polygon / Point converted to Raster		
Tree DBH	Field Measurement	/	/	28.67 (7.08)

Mean and standard deviation illustrate range of values captured among sites. Final analysis used standardized predictors.

Table 3. Summary of final predictive models of mountain pine beetle sex ratio skew for **all beetles** (larvae + adults) ranked according to their $\Delta AICc$ values

Model no.	Model code	Model description	K	$\Delta AICc$	w	Σw	Log-Lik
1 ^a	~ DBH + MaxSummer + MaxFall + MeanSpring + MAT + Latitude + Longitude	Tree size + weather + location + climate	8	0.00	0.44	0.44	-1527.79
2	~ DBH + YiO + MaxSummer + MaxFall + MeanSpring + Latitude	Tree size + year in outbreak + weather + location	7	0.38	0.36	0.80	-1528.99
3	~ DBH + YiO + MaxSummer + MaxFall + MeanSpring + Latitude + MAT + CMD	Tree size + year in outbreak + weather + location + climate	9	3.66	0.07	0.87	-1528.61
4	~ DBH + YiO + MaxSummer + MaxFall + MinWinter + MeanSpring	Tree size + year in outbreak + weather	7	3.90	0.06	0.93	-1530.75
5	~ DBH + YiO + MaxSummer + MaxFall + MeanSpring + MinWinter + MAT + CMD	Tree size + year in outbreak + weather + climate	9	4.39	0.05	0.98	-1528.97
6	~ DBH + YiO	Tree size + year in outbreak	3	8.36	0.01	0.99	-1536.99
7	~ DBH + YiO + Latitude	Tree size + year in outbreak + location	4	9.19	0.00	1.00	-1536.41
8	~ DBH + YiO + MAT + CMD	Tree size + year in outbreak + climate	5	10.19	0.00	1.00	-1535.90
9	~ DBH + YiO + Latitude + MAT + CMD	Tree size + year in outbreak + location + climate	6	11.15	0.00	1.00	-1535.38
10	~1	Intercept only	1	13.57	0.00	1.00	-1541.60

K indicates the number of parameters in each model, w indicates model AICc weights, Σw indicates cumulative weights, and Log-Lik indicates the log-likelihood of the model. Details on predictor variables used to construct models are found in Table 2. Coefficient plots for the final selected model are shown in Fig. 5A.

^aSelected using forward selection from all possible predictors (Table 2).

size-categories based on DBH: (1) <20 cm; (2) 20–25 cm; (3) 26–30 cm; (4) 31–35 cm; (5) 36–40 cm; and (6) >40 cm. Within each category within each stand, the mean DBH of included trees was used for prediction. Reclassifying our data in this way yielded 124 unique size-category/stand classes (FC.1_1, FC.1_2, etc.). Ninety-nine and 123 classes were available when examining the adult and larvae only models, respectively. We predicted sex ratio expressed as a percentage using these data. Performance was assessed using the adjusted R^2 of a linear regression of all observed values on all predicted values (Piñeiro et al. 2008). Regression models were built using the *lm* function in R.

We also used a leave one out cross-validation approach (LOOCV) to assess the models' out of sample prediction at the level of tree size categories. Models were repeatedly fit using individual-level data excluding individuals from a single stand. Using this model, we then predicted tree-level sex ratio skew in the excluded stand. Performance was assessed using the average of one minus the root mean squared error (RMSE) between observed and predicted sex ratio at the tree size category level (i.e., mean(1-RMSE)) over all site exclusions. 1-RMSE has a maximum value of 1 that indicates perfect concordance; 0 indicates no concordance.

Results

Spatial Temporal Sex Ratio Variation

Live adult and larval beetles were found throughout our sampling period (Fig. 2; Table 1). Our analysis included half the range examined in Amman and Cole (1983) (0–6 yr), and a broad range in tree diameters ($\bar{x}$ = 28.67 cm; SD = 7.09 cm; Table 2). In accordance with previously described 2:1 ratios of females to males (67% ♀; Reid 1958), we consistently found more female than male beetles in all stands with a global average ratio of 1.79:1 (64.2% ♀; SD = 7.3%). The global average ratio for larvae was 1.76:1 (63.8% ♀; SD = 9.5%) and for adults was 2:1 (66.6% ♀; SD = 11.7%).

Indeed, in examining all beetles (larvae + adults), life-stage (adult vs. larvae) was a significant determinant of beetle sex and a greater proportion of adults than larvae were female ($\chi^2 = 5.70$, $P = 0.016$). Among sites, the range of sex ratio ranged from 0.8:1 to 5.5:1 for all beetles (larvae + adults), 0.77:1 to 3.3:1 for larvae (excluding Cypress Hills that contained a single female larvae), and from 0.8:1 to 22:1 for adults. This exceptional value of 22:1 was found in a stand in Fox Creek (FC.1; Fig. 3) where of the 23 adult beetles collected in January 2008, 22 of them (95%) were female. In contrast, the larval ratio at this site was 2.1:1 ($n = 65$).

Sex ratio skew in adult beetles varies spatially at the scale of our study area. The site from which samples were taken (Table 2, Fig. 3) was a significant predictor of sex ratio for adult beetles ($F = 1.97$, $df = 11$, $P = 0.01$) in logistic regression. When considering all beetles together, the relationship was nonsignificant ($F = 1.70$, $df = 11$, $P = 0.06$) and largely driven by the lack of any relationship between larval sex ratio and site ($F = 1.24$, $df = 11$, $P = 0.26$). Geographic location (latitude and longitude) explained adult beetle sex ratio. When considered individually, longitude appeared to have a larger effect. Latitude was only significant ($P = 0.002$) after longitude ($P < 0.001$) entered the model. No interaction effect between latitude and longitude was detected ($P = 0.59$). The probability of finding adult female beetles generally increases from west to east and from south to north, that is, along the main axes of expansion. This spatial relationship remains significant even after removing the most eastern site from the analysis (Cypress Hills; Fig. 3). Neither latitude nor longitude helped explain sex ratio skew when considering larvae. We found no effect of collection date (i.e., Fig. 2).

Hypothesis Testing

We identified a significant interaction effect between beetle life-stage (i.e., larva vs. adult) and DBH ($P = 0.02$) as well as between life-stage and YiO ($P = 0.01$). We therefore constructed models considering all beetles (larvae + adults), adults, and larvae separately. Tree

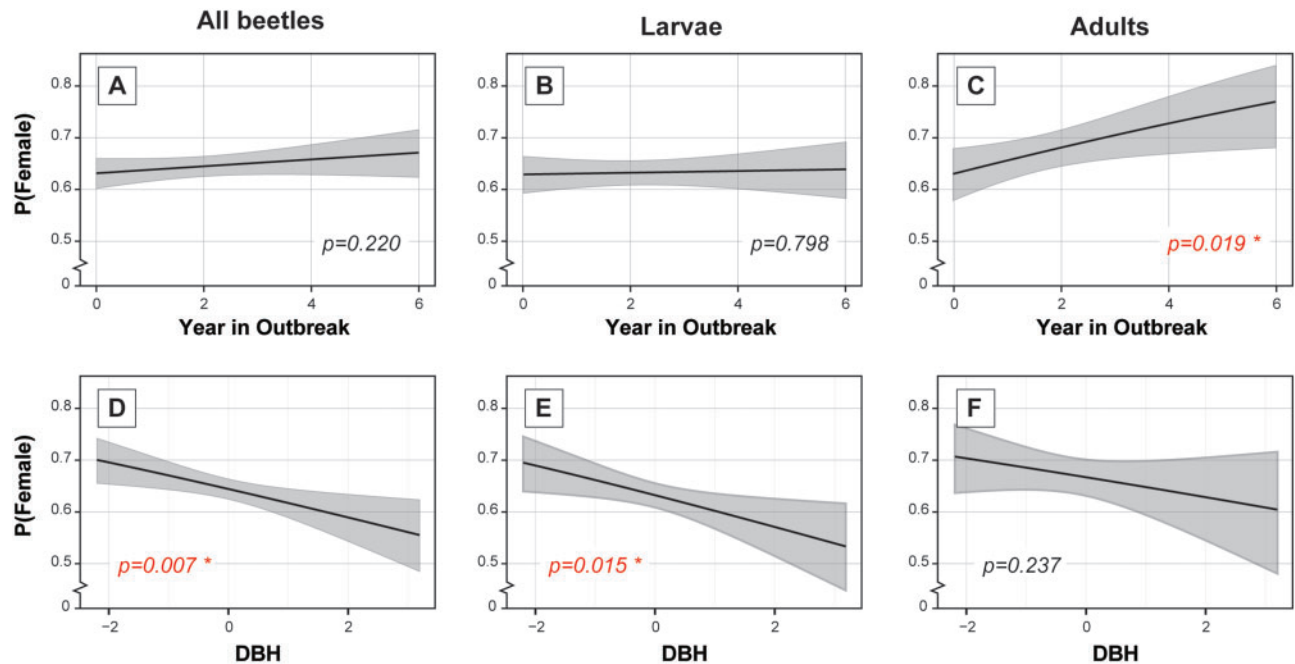


Fig. 4. Predicted influences of Year in Outbreak and DBH on the probability that a selected beetle is female, separated into all beetles (larvae + adults; panels **A** & **D**), larval beetles only (Panels **B** & **E**), and adult beetles only (Panels **C** & **F**). *P*-values in each panel indicate significance of the predictor for that subset of data. Contrast with Figure 1.

diameter (DBH) significantly influences beetle sex ratio in all beetles ($P = 0.007$; Fig. 4D) and larvae ($P = 0.014$; Fig. 4E). In accordance with previous work (Fig. 1), we found that the smaller the tree, the higher the probability that a beetle was female. Year in outbreak (YiO) significantly influenced sex ratio in adults ($P = 0.02$; Fig. 4A) but not in larvae ($P = 0.79$; Fig. 4B) or all beetles together ($P = 0.20$; Fig. 4F). No significant interaction between YiO and DBH was detected, contrary to the hypothesis implicit in Fig. 1. However, there was a significant interaction between DBH and Site in larvae ($P = 0.024$), suggesting that the relationship between tree size and larval sex ratio may be further influenced by spatial context. To summarize, populations of both larval and adult beetles were female biased but adults were more biased. Larval sex ratio skew was influenced by tree diameter whereas adult beetles were significantly affected by the number of years that a site has been undergoing an outbreak. These results provide partial, but not complete, support for the model proposed by Amman and Cole (1983) (Fig. 1).

Predictive Models

When considering all beetles (larvae + adults), the best model (model 1, Table 3) included tree diameter, maximum summer and fall temperatures, fall temperatures, mean annual temperature, and as well as latitude and longitude. A second model with equal support ($\Delta\text{AICc} < 2$; Burnham and Anderson, 2002) included tree diameter, year in outbreak, latitude mean spring, and maximum summer and fall temperatures (Fig. 5A). Using an information theoretic approach, we are unable to distinguish between these models; both are potentially good models for predicting beetle sex. However, on the basis of parsimony (fewer parameters), interpretability (i.e., both DBH and YiO are included), minimization of correlations (e.g., latitude and longitude), and concordance with the larval model described below, we chose to use model 2 for prediction (Table 3). When assessing the degree of support for model 2 without

considering model 1 relative to the other candidate models, it is clearly the best model ($\Delta\text{AICc} = 0$; AICc weight = 0.65).

The best larval model included the same variables as the adult model 2 (Table 4; Fig. 5B). This model was identical to the model chosen using step-wise selection and was strongly supported by its AICc weight ($w = 0.69$; Table 4).

When examining adult beetles, the model identified using step-wise selection was best. This model included latitude, longitude, and maximum fall temperature (Table 5; Fig. 5C) and was strongly supported by its AICc weight ($w = 0.96$; Table 5). None of the other candidate models was supported. Removal of the extremely skewed adult site (Fox Creek; 22/23 ♀) resulted in preference for a similar model that replaced latitude with minimum winter temperature and differed with respect to ΔAICc and $\Delta\text{Log-Likelihood}$ by less than 1. Thus, we do not consider Fox Creek as a source of bias in our model. We include three short R-scripts to translate new field data into predicted estimates of brood production and sex ratio skew using the best model for each grouping (Supp. Material 1 [online only]).

Model Performance

Model performance was sensitive to the number of individuals included in a size category (Fig. 6). For both the all beetle and larval models (not shown), the AdjR^2 of the regression model between observed and predicted values increases with the number of individuals used to estimate tree-level sex ratio. The adult model did not show further improvement (above an already high AdjR^2) with sample sizes above 16 individuals per size-category / stand combination (not shown). However, for all cases, as we increased the threshold value, there were fewer size-category / stand combinations available for the comparison of observed and predicted values. We sought to identify a balance between the number of individuals included per size-category and the number of size-category / stand combinations to provide a meaningful evaluation of model performance. For example,

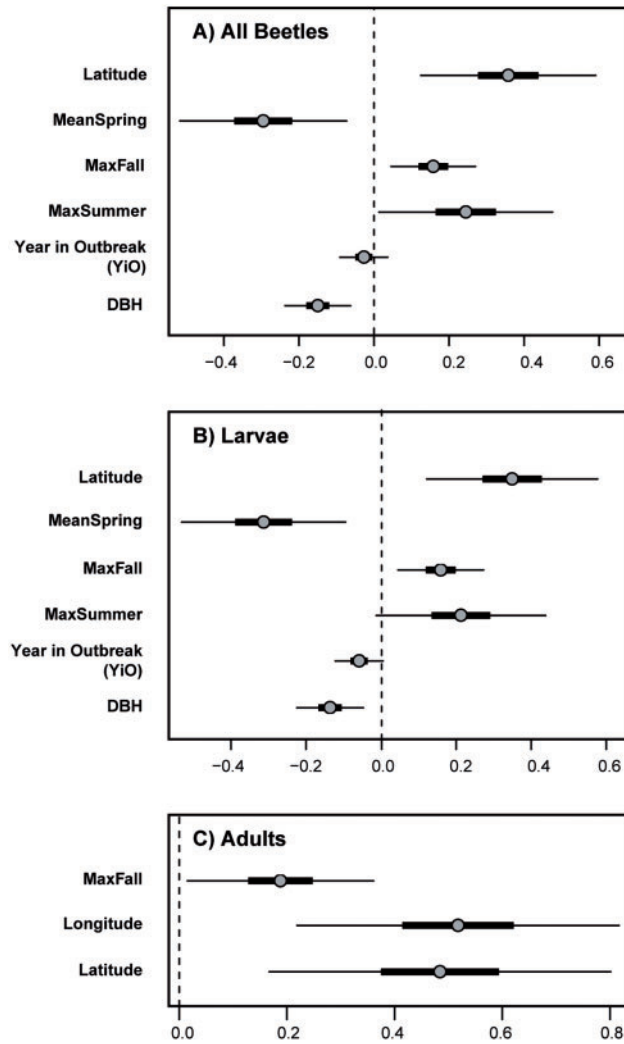


Fig. 5. Individual coefficient plots for the best model predicting probability of female occurrence selected using a minimized AICc for: (A) all beetles (larvae + adults), (B) larvae, and (C) adults. Seasonal variables refer to temperature. Horizontal bars represent 1 and 2 standard deviations of the coefficient estimate.

Table 4. Summary of final predictive models of mountain pine beetle sex ratio skew for **larval beetles** ranked according to their Δ AICc values

Model no.	Model code	Model description	<i>K</i>	Δ AICc	<i>w</i>	Σw	Log-Lik
1 ^a	~ DBH + YiO + MaxSummer + MaxFall + MeanSpring + Latitude	Tree size + year in outbreak + weather + location	7	0.00	0.69	0.69	-1070.47
2	~ DBH + YiO + MaxSummer + MaxFall + MeanSpring + Latitude + MAT + CMD	Tree size + year in outbreak + weather + location + climate	9	2.47	0.20	0.90	-1069.68
3	~ DBH + YiO + MAT + CMD	Tree size + year in outbreak + climate	5	6.92	0.02	0.92	-1075.95
4	~ DBH + YiO	Tree size + year in outbreak	3	6.97	0.02	0.94	-1077.98
5	~ DBH + YiO + MaxSummer + MaxFall + MinWinter + MeanSpring	Tree size + year in outbreak + weather	7	7.46	0.02	0.96	-1074.20
6	~ DBH + YiO + Latitude	Tree size + year in outbreak + location	4	7.79	0.01	0.97	-1077.39
7	~ DBH + YiO + Latitude + MAT + CMD	Tree size + year in outbreak + location + climate	6	7.91	0.01	0.98	-1075.43
8	~ DBH + YiO + MaxSummer + MaxFall + MeanSpring + MinWinter + MAT + CMD	Tree size + year in outbreak + weather + climate	9	8.68	0.01	0.99	-1072.79
9	~1	Intercept only	1	9.04	0.00	1.00	-1081.02

K indicates the number of parameters in each model, *w* indicates model AICc weights, Σw indicates cumulative weights, and Log-Lik indicates the log-likelihood of the model. Details on predictor variables used to construct models are found in Table 2. Coefficient plots for the final selected model are shown in Fig. 5B.

^aThis model was the same as that selected using automatic forward selection

when considering all data, if we were to be very strict and only included size categories with more than 50 individuals, only 5 size-category / stand combinations would be available for prediction. In contrast, if we were to (erroneously) assume that 5 individuals were sufficient to estimate tree-level sex ratio, 109 combinations would be available. The number of individuals per tree size-category also differed among adult, larval, and combined datasets. With these considerations in mind, we chose to use tree-size categories that contained at least 40 individuals when considering all data, 16 individuals for adults, and 30 individuals for larvae. It is important to bear in mind that the resulting values represent our best statistical estimates of tree level sex ratio, rather than the true values.

There was significant correlation between predicted and observed values for all models (Fig. 7). The adult model ($\text{AdjR}^2 = 0.81$, $t = 6.561$, $P < 0.001$, $df = 9$) performed better than the larvae only model ($\text{AdjR}^2 = 0.58$, $t = 3.443$, $P = 0.011$, $df = 7$), which in turn performed better than the model that used all data ($\text{AdjR}^2 = 0.45$, $t = 3.288$, $P = 0.007$, $df = 11$). Models tended to over predict sex ratio in non-female-skewed populations ($P(\varphi) < 0.5$) and under predict skew in very strongly female-skewed populations ($P(\varphi) > 0.8$). Performance was improved by removing the size-category / site combinations with extreme skews (e.g., sites FC.1_3, KY.2_2; Table 1) in the all data model (i.e., AdjR^2 increased from 0.45 to 0.75). However, adjusted R^2 was slightly reduced when extreme sites were removed from the adult and larval models. All models remained significant after removal of these sites. Leave one out cross validation also indicated that our models perform reasonably well. The mean performance (i.e., 1-RMSE) for larval, adult, and all beetle (larvae + adults) models was 0.85 (SD = 0.07; $n = 34$), 0.81 (SD = 0.11; $n = 32$), and 0.88 (SD = 0.07; $n = 34$), respectively.

Discussion

Sex ratio skew exists in both larval and adult mountain pine beetle, but is more pronounced in adults. This suggests that sex ratio skew in the mountain pine beetle likely originates during development, due to a higher rate of male mortality relative to females. Had sex ratio skew originated strictly at the time of oviposition, we would expect to see a similar sex ratio skew between larvae and adults that varied equivalently in space. However, some interactions of the two

Table 5. Summary of final predictive models of mountain pine beetle sex ratio skew for **adult beetles** ranked according to their $\Delta AICc$ values

Model no.	Model code	Model description	K	$\Delta AICc$	w	Σw	Log-Lik
1 ^a	~ MaxFall + Latitude + Longitude	Weather + location	4	0.00	0.96	0.96	-449.60
2	~ DBH + YiO	Tree size + year in outbreak	3	8.74	0.01	0.97	-454.99
3	~ DBH + YiO + MaxSummer + MaxFall + MinWinter + MeanSpring	Tree size + year in outbreak + weather	7	8.78	0.01	0.99	-450.94
4	~ DBH + YiO + Latitude	Tree size + year in outbreak + location	4	10.68	0.00	0.99	-454.94
5	~ DBH + YiO + MAT + CMD	Tree size + year in outbreak + climate	5	12.07	0.00	0.99	-454.62
6	~ DBH + YiO + MaxSummer + MaxFall + MeanSpring + MinWinter + MAT + CMD	Tree size + year in outbreak + weather + climate	9	12.21	0.00	1.00	-450.61
7	~1	Intercept only	1	12.53	0.00	1.00	-458.89
8	~ DBH + YiO + MaxSummer + MaxFall + MeanSpring + Latitude	Tree size + year in outbreak + weather + location	7	12.81	0.00	1.00	-452.96
9	~ DBH + YiO + Latitude + MAT + CMD	Tree size + year in outbreak + location + climate	6	13.87	0.00	1.00	-454.51
10	~ DBH + YiO + MaxSummer + MaxFall + MeanSpring + Latitude + MAT + CMD	Tree size + year in outbreak + weather + location + climate	9	15.76	0.00	1.00	-452.38

K indicates the number of parameters in each model, w indicates model AICc weights, Σw indicates cumulative weights, and Log-Lik indicates the log-likelihood of the model. Details on predictor variables used to construct models are found in Table 2. Coefficient plots for the final selected model are shown in Fig. 5C.

^aSelected using forward selection from all possible predictors (Table 2).

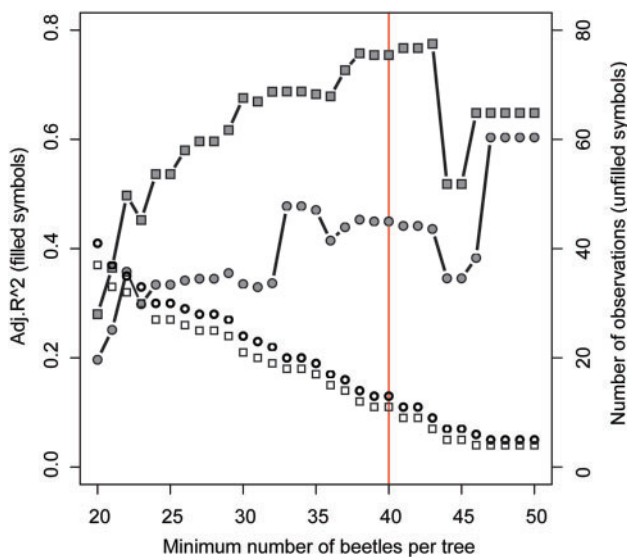


Fig. 6. Model performance varies with sample size within tree age-categories. When considering the selected model for all beetles (larvae + adults), the adjusted R^2 describing the regression between predicted and observed values changes with the threshold number of beetles used to estimate tree-level sex ratio. Filled symbols show how adjusted R^2 changes with increasing minimum threshold for the number of beetles (left-hand vertical axis). Unfilled symbols indicate the consequent number of size-class / site categories (n in regression models) used to fit the regression models (right-hand vertical axis). Circles show results for all sites, and squares show the same patterns after the very strongly skewed tree size-class site combination FC.1.3 ($p(\varnothing) = 0.9$) and the nonskewed combination KY.2.2 ($p(\varnothing) = 0.45$) were removed from analysis. Vertical (red) line indicates sample size chosen for assessing model performance and illustrated in Fig. 7A. See Fig. 7 and text for details.

may also be operating as these two processes are not necessarily mutually exclusive. According to our results and those of others (Lachowsky and Reid 2014), sex ratio skew in the mountain pine beetle is the consequence of interactions between male beetles and their environment and not due to a response of adult females during

oviposition. Based on our results, as the mountain pine beetle outbreak continues to move eastward and northward into novel habitats with generally smaller trees and colder temperatures, one might expect to see even greater female-biased sex ratios. This hypothesis remains to be verified.

Lachowsky and Reid (2014) demonstrated at a single location that sex ratio skew in the mountain pine beetle was due to differential mortality between sexes. At a broader, regional scale we also found evidence for differential mortality, specifically driven by tree size, weather, location and local outbreak history (Fig. 5). Smaller trees result in relatively fewer males and hence increased female bias in adult populations (e.g., Fig. 4). Weather, specifically warmer summer and colder winter temperatures, is expected to disfavor male survival (Safranyik et al. 1999). In the sexually dimorphic mountain pine beetle, males are generally smaller, have fewer fat reserves, and are less tolerant to environmental stress including freezing and desiccation (Safranyik et al. 1999, Régnière and Bentz 2007, Reid and Purcell 2011). Although we identified an influence of geographic location on mountain pine beetle sex ratio, we cannot be certain what processes are captured by this correlation. Geographic location can serve as a surrogate for many other nonmeasured environmental variables such as local weather or climate, forest structure (composition and configuration), or edaphic factors. The increase in female-biased sex ratio skew from the historical range in the west into new areas in the east and north could be a result of different climate, differences in forest fragmentation and mountain pine beetle population connectivity (Safranyik et al. 2010), different host species availability (e.g., hybrid lodgepole-jack pine and jack pine (Cullingham et al. 2011), or naïve trees with reduced defensive capacity (Cudmore et al. 2010). Further work will be required to refine our understanding of the precise mechanisms affecting variation in sex ratio along these expansion axes.

The scale of spatial variation in mountain pine beetle sex ratio differed between larvae and adults. Adult sex ratio was influenced by broad-scale geographic variation (e.g., sites). In contrast, larvae appear to be affected by both broad-scale processes and finer scale, within site processes linked to individual tree diameter. The relationship between DBH, male mortality, and female-biased sex ratio is

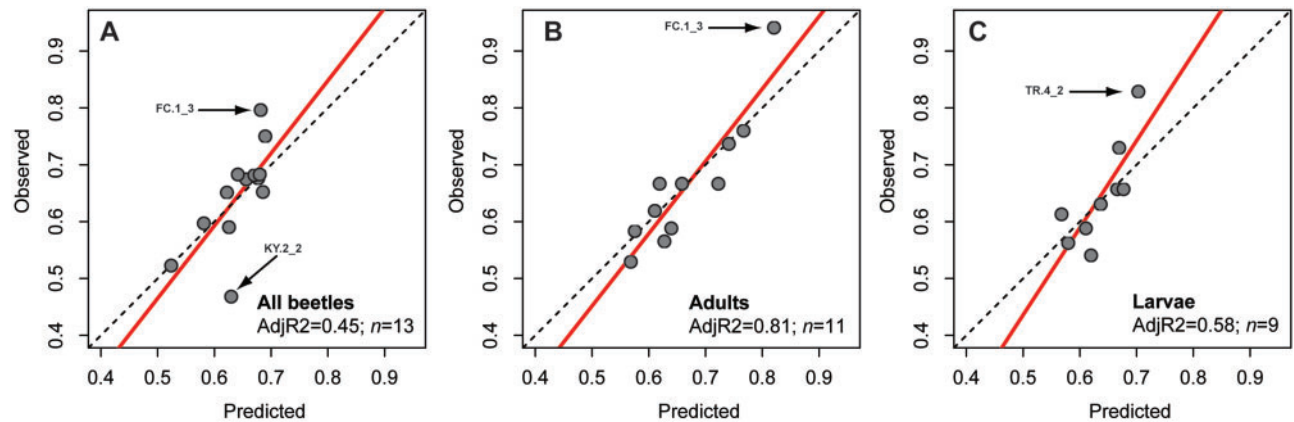


Fig. 7. Performance of selected predictive models for (A) all data; (B) adult beetles, and (C) larval beetles. Solid red lines indicate best fit regression line fit using ordinary least squares. Dashed line represents 1-1 line. Arrows indicate size-class / site combinations with very strongly or weakly female skewed populations. Labels indicate the site (first three characters; see Table 1) and the size class (number following the underscore). Model fit was only improved by removal of these sites in the all beetle model. Details of each regression model used to assess correspondence between predicted and observed values can be found in the text.

considered well understood has been previously described (Cole et al. 1976); males are consistently more vulnerable in smaller trees due to environmental stress. However, tree host nutritional status may also play a role in differential beetle mortality in smaller trees (Bentz and Mullins 1999), as phloem-feeding bark beetles are typically nutrition (e.g., Nitrogen) limited (Ayres et al. 2000). Adult beetles were influenced by YiO, but larvae were not (Fig. 4). The inverse was true with regards to tree diameter (Fig. 4). Together, these results further indicate that different processes are involved in determining beetle sex ratio at different life-stages.

The differences between determinants of larval and adult sex ratios is intriguing, as one would expect that that patterns established during larval development would remain in the adult stage. One interpretation is that due to spatial dependence (Dale and Fortin 2014), the predictors Site and YiO act as proxies for other spatial variables such as climate or weather such that areas that are more climatically suitable are attacked earlier in the outbreak. While differential larval mortality is affected by local, within tree conditions (DBH) the larval sex ratio skew is then further modified in later developmental stages by broader scale climate-related processes. YiO may also capture the consequences of an increase in population density. One might expect that as a regional outbreak progresses and attacks are successful, within-tree population density increases, resources are depleted, and phloem dries out. The resulting within-tree competition and stress reduce male survival (Safranyik et al. 1999). The precise physiological mechanisms driving these different responses in larval and adult beetles are uncertain but are likely related to important differences in terms of lipid content and consequent cold tolerance, as there are between sexes (Lombardero et al. 2000).

Revising an Existing Model

Previous work has also shown an increase in the proportion of females during early outbreak development in small trees and late in outbreak development in large trees (Amman and Cole 1983, Fig. 1). We did not observe such an increase in male survival in large trees during the early stages of an outbreak as suggested by Fig. 1, nor did we identify any sort of interaction between outbreak development (YiO) and DBH (Amman and Cole 1983). Further, although adult female bias increased monotonically (Fig. 4C), we do not see the uni-modal pattern in sex ratio shown in Fig. 1. Because the YiO values from our sites cover only six years, it seems that we are

limited to the left half of Fig. 1 (years 0-6). Our results based on YiO between 0 and 6 years support a portion of Amman and Cole's conceptual model of sex ratio variation through time. However, the lack of an interaction effect between DBH and YiO suggests that this model may require revision. Further sampling of multiple locations through time and in different environmental contexts (e.g., temperature, outbreak stage, forest structure) is needed to refine this relationship and its importance to beetle population dynamics. Transects across the leading edge of the outbreaks should also be examined in greater detail.

Predictive Models

The best models for predicting beetle sex ratio differed between larval and adult beetles (Fig. 5). For larvae, models that included only DBH and YiO were far inferior to those that also included information on local environmental context (i.e., weather and location; Tables 3, 4, Fig. 5). The inclusion of weather and location variables reflects the influence of local thermal regimes that can contribute to beetle stress, desiccation, and mortality (Bentz et al. 2010). YiO was also included and may represent a proxy for other predictors (e.g., climate) that were not explicitly measured. Though no single weather variable we examined demonstrated a correlation with YiO > 0.52 , suggesting that another process, possibly competition due to increased population densities, may be captured by this predictor. Also, we used a linear model to describe the relationship between sex ratio and YiO (see Fig. 4C) whereas a longer term original prediction involves a unimodal relationship (Fig. 1). Our predictive models are thus limited to situations where a linear trend in sex ratio in response to YiO is present.

Spatial variation in adult beetle sex ratio was best predicted using a simple model that included latitude, longitude, and maximum fall temperature. The difference between the adult and larval models suggests to us that there are additive or cross-scale dynamics involved in determining sex ratio skew during development. It is curious that tree diameter had no influence, as we would expect the conditions in which the larvae developed to influence adult sex ratio skew. Strict data filtering prior to analysis assures us that these adults were indeed of the current year's generation (Fig. 2). Nonetheless, these adults may represent a biased subset in some way that we are not aware of such that the full population of adults to emerge in the spring would exhibit the sex ratio skew of the larval population.

All models performed well and showed scatter and regression lines very close to the 1:1 line (Fig. 7). The adjusted R^2 indicates that the model for adult beetles performs the best despite having fewer adults than larvae with which to estimate tree-level sex ratio (Table 1). The larval model performs less well possibly because there is more heterogeneity and finer scale processes operating within the larval category (e.g., developmental stages) than is accounted for in this analysis. Forest managers will have the greatest chance to correctly predict tree-level mountain pine beetle sex ratio using our adults-only model (see Suppl. Material [online only]). Within a restricted range of potential sex ratio (i.e., $0.5 < \text{sex ratio} < 0.8$) the all beetle model is also expected to perform well.

Opportunities remain to improve models to predict mountain pine beetle sex ratio skew. Future work should focus on collecting detailed environmental information along with the presence and absence of male and female beetles in areas showing either very highly or very weakly (or non-) female biased populations as these are the contexts about which we currently know the least. Higher number of individuals per tree would also be useful to improve our estimates of the real tree-level sex ratio skew and could lend greater weight to assessments of model performance.

Other Potential Determinants

Sib-mating has been shown to alter sex ratios from parity (Hamilton 1967). Pre-emergence sib-mating including polygamy has been demonstrated in the mountain pine beetle (Bleiker et al. 2013, Janes et al. 2015), and we cannot completely discount its potential influence on observed sex ratios. However, given the generally low rates of observed pre-emergence mating in mountain pine beetle populations (3–12%; Bleiker et al. 2013) we do not expect it to exert a strong influence within epidemic populations. It is also difficult to determine whether biased sex ratios are the result of sib-mating, differential mortality, differential dispersal of females (Safranyik and Wilson 2006). Differential dispersal losses could also affect sex ratio in the fall although as far as we know there is no evidence for this in the mountain pine beetle (Reid 1958). We also do not know if the frequency of sib-mating varies in response to spatial context. Although we do not know what sort of bias (if any) was present initially, our results and those of recent studies (Lachowsky and Reid 2014, Lachowsky et al. 2015) suggest that sex ratio skew likely does not originate at oviposition, but instead arises gradually as a result of differential male mortality over the course of development. Future studies dedicated to monitoring population sex ratio from egg through all developmental stages to adult in varying environmental conditions (e.g., climate, outbreak stage, and nearer the center of the mountain pine beetle's historical range) is needed to further clarify this issue.

Current aspatial management models of mountain pine beetle outbreak dynamics (Safranyik et al. 1999) consider the processes that determine sex ratio to be temporally constant (i.e., stationary; Dale and Fortin 2014) during outbreaks. In demonstrating that sex ratio skew most likely arises due to male mortality during development rather than due to initial female-bias at oviposition (potentially in response to outbreak conditions), we conditionally support this assumption and posit that sex ratio skew in the mountain pine beetle is more likely a consequence of the outbreak than a cause. Spatial variation in the realization of these processes, which itself may or may not be stationary, nonetheless creates heterogeneity in rates of male mortality and sex ratio at the site and tree levels. Current monitoring and forecasting could be improved by examining how spatial

variation in sex ratio affects brood productivity, population dynamics, and stand susceptibility.

Theoretical predictions about sex ratio skew are usually expressed using evolutionary time scales. However, in the context of irruptive systems in heterogeneous environments, ecological processes at finer temporal scales are also important to consider. Interactions among environmental context, density dependence, temperature, and differential susceptibility between sexes can lead to important spatial and temporal variation in sex ratio that in turn affect effective beetle population sizes and population growth rates. Unravelling the complex and sometimes cross-scale relationships between spatial context and sex ratio, as well as understanding their consequences, is important for the management of ecological systems that are being faced with novel environments due to rapid changes in land use and climate.

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